TSM 098

Homework 1

Number structure, signs, and fraction reduction

Topics 1-4 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

Factors and signs

1. Compute without a calculator. Use a related multiplication fact to explain each division.

$7 \cdot 8, \quad 9 \cdot 6, \quad 56 \div 7, \quad 54 \div 9$

2. For each number, identify which of 2, 3, 5, 9, and 10 divide it. Explain using divisibility rules.

$126, \quad 240, \quad 315$

3. Write each prime factorization.

$72, \quad 90, \quad 140$

4. Explain why 1 is not prime and why 49 is not prime.

5. Evaluate each sum or difference.

$-8 + 13, \quad 6 - 11, \quad -4 - 9, \quad -7 + 7$

6. Evaluate each expression. Explain the sign of each result.

$(-6)(-4), \quad (-3)(5), \quad \frac{-24}{6}, \quad \frac{-35}{-7}$

7. Evaluate or state that the expression is undefined. Explain the role of zero.

$\frac{0}{8}, \quad \frac{8}{0}, \quad \frac{0}{0}, \quad 0(-5)$

8. Find two integers with each specified sum and product: (a) sum 1, product -12; (b) sum -7, product 12; (c) sum 7, product 12.

Reduce each fraction completely

9. Simplify and show common factors.

$\frac{42}{56}$

10. Simplify and show common factors.

$\frac{-54}{81}$

11. Simplify and show common factors.

$\frac{84}{126}$

12. Simplify and show common factors.

$\frac{-45}{-75}$

13. Simplify and show common factors.

$\frac{120}{168}$

14. Simplify and show common factors.

$\frac{-36}{48}$

Check structure

15. Fill in the missing numerator and explain why the fractions are equivalent.

$\frac{3}{5} = \frac{n}{40}$

16. A student reduces 12/18 to 2/8 by deleting the digit 1 from the numerator and denominator. Explain why this is invalid; reduce 12/18 correctly.

17. Without decimals, decide which is larger: -3/4 or -2/3. Use equivalent fractions and explain the effect of the negative signs.